

```

/* =====
TRANSACTION NOTIFICATION
Automatically creates context-specific notification after
every successful transaction
===== */

```

```

CREATE OR ALTER TRIGGER trg_Transactions_Notification
ON Transactions
AFTER INSERT
AS
BEGIN

```

```

    SET NOCOUNT ON;

```

```

    INSERT INTO Notifications
    (

```

```

        CustomerID,
        NotificationType,
        Subject,
        Message,
        DeliveryChannel,
        SentDate,
        DeliveryStatus
    )

```

```

    SELECT

```

```

        C.CustomerID,

```

```

        -- Notification type
        'Transaction',

```

```

        -- Transaction-specific subject

```

```

    CASE

```

```

        WHEN I.TransactionCategory = 'Cash Deposit'

```

```

            THEN 'Cash Deposit Successful'

```

```

        WHEN I.TransactionCategory = 'ATM Deposit'

```

```

            THEN 'ATM Deposit Successful'

```

```

        WHEN I.TransactionCategory = 'Cheque Deposit'

```

```

            THEN 'Cheque Deposit Successful'

```

```

        WHEN I.TransactionCategory = 'Interest Credit'

```

```

            THEN 'Interest Credited'

```

```

        WHEN I.TransactionCategory = 'Cash Withdrawal'

```

```

            THEN 'Cash Withdrawal Successful'

```

```

        WHEN I.TransactionCategory = 'ATM Withdrawal'

```

```

            THEN 'ATM Withdrawal Successful'

```

```

        WHEN I.TransactionCategory = 'Cheque Clearance'

```

```

            THEN 'Cheque Clearance Successful'

```

```

        WHEN I.TransactionCategory = 'Charges'

```

```

            THEN 'Bank Charge Applied'

```

```

        WHEN I.TransactionCategory = 'FD Booking'

```

```

            THEN 'Fixed Deposit Booked'

```

```

        WHEN I.TransactionCategory = 'RD Booking'

```

```

            THEN 'Recurring Deposit Booked'

```

```

        WHEN I.TransactionCategory = 'EMI Payment'

```

```

            THEN 'EMI Payment Successful'

```

```

        WHEN I.TransactionCategory IN ('UPI', 'IMPS', 'NEFT', 'RTGS')

```

```

            THEN I.TransactionCategory + ' Transfer Successful'

```

```

        ELSE 'Transaction Successful'

```

```

    END,

```

```

    -- Transaction-specific message

```

```

    CASE

```

```

        WHEN I.TransactionCategory = 'Cash Deposit'

```

```

            THEN 'Cash deposit of Rs. ' + CAST(I.Amount AS VARCHAR(30))
            + ' completed successfully. Reference: ' + I.TransactionReferenceNumber

```

```

        WHEN I.TransactionCategory = 'ATM Deposit'

```

```

            THEN 'ATM deposit of Rs. ' + CAST(I.Amount AS VARCHAR(30))
            + ' completed successfully. Reference: ' + I.TransactionReferenceNumber

```

```

        WHEN I.TransactionCategory = 'Cheque Deposit'

```

```

            THEN 'Cheque deposit of Rs. ' + CAST(I.Amount AS VARCHAR(30))
            + ' recorded successfully. Reference: ' + I.TransactionReferenceNumber

```

```

        WHEN I.TransactionCategory = 'Interest Credit'

```

```

            THEN 'Interest of Rs. ' + CAST(I.Amount AS VARCHAR(30))
            + ' credited successfully. Reference: ' + I.TransactionReferenceNumber

```

```

        WHEN I.TransactionCategory = 'Cash Withdrawal'

```

```

            THEN 'Cash withdrawal of Rs. ' + CAST(I.Amount AS VARCHAR(30))
            + ' completed successfully. Reference: ' + I.TransactionReferenceNumber

```

```

        WHEN I.TransactionCategory = 'ATM Withdrawal'

```

```

            THEN 'ATM withdrawal of Rs. ' + CAST(I.Amount AS VARCHAR(30))
            + ' completed successfully. Reference: ' + I.TransactionReferenceNumber

```

```

        WHEN I.TransactionCategory = 'Cheque Clearance'

```

```

            THEN 'Cheque clearance of Rs. ' + CAST(I.Amount AS VARCHAR(30))
            + ' completed successfully. Reference: ' + I.TransactionReferenceNumber

```

```

        WHEN I.TransactionCategory = 'Charges'

```

```

            THEN 'Bank charge of Rs. ' + CAST(I.Amount AS VARCHAR(30))
            + ' has been applied. Reference: ' + I.TransactionReferenceNumber

```

```

        WHEN I.TransactionCategory = 'FD Booking'

```

```

            THEN 'Fixed Deposit of Rs. ' + CAST(I.Amount AS VARCHAR(30))
            + ' booked successfully. Reference: ' + I.TransactionReferenceNumber

```

```

        WHEN I.TransactionCategory = 'RD Booking'

```

```

            THEN 'Recurring Deposit of Rs. ' + CAST(I.Amount AS VARCHAR(30))
            + ' booked successfully. Reference: ' + I.TransactionReferenceNumber

```

```

        WHEN I.TransactionCategory = 'EMI Payment'

```

```

            THEN 'EMI payment of Rs. ' + CAST(I.Amount AS VARCHAR(30))
            + ' completed successfully. Reference: ' + I.TransactionReferenceNumber

```

```

        WHEN I.TransactionCategory IN ('UPI', 'IMPS', 'NEFT', 'RTGS')

```

```

            AND I.TransactionType = 'Debit'

```

```

            THEN I.TransactionCategory + ' transfer of Rs. '
            + CAST(I.Amount AS VARCHAR(30))
            + ' completed successfully. Reference: '
            + I.TransactionReferenceNumber

```

```

        WHEN I.TransactionCategory IN ('UPI', 'IMPS', 'NEFT', 'RTGS')

```

```

            AND I.TransactionType = 'Credit'

```

```

            THEN 'Rs. ' + CAST(I.Amount AS VARCHAR(30))
            + ' received through ' + I.TransactionCategory
            + '. Reference: ' + I.TransactionReferenceNumber

```

```

        ELSE 'Transaction of Rs. ' + CAST(I.Amount AS VARCHAR(30))

```

```

            + ' completed successfully. Reference: '
            + I.TransactionReferenceNumber

```

```

    END,

```

```

    -- Current project simulation uses SMS

```

```

    'SMS',

```

```

    SYSDATETIME(),

```

```

    -- Notification considered successfully sent

```

```

    'Sent'

```

```

    FROM inserted I

```

```

    INNER JOIN Accounts A

```

```

        ON I.AccountID = A.AccountID

```

```

    INNER JOIN Customers C

```

```

        ON A.CustomerID = C.CustomerID

```

```

    WHERE I.TransactionStatus = 'Success';

```

```

END;

```

```

GO

```

110 %

Results Messages

	NotificationID	CustomerID	NotificationType	Subject	Message	DeliveryChannel	SentDate	DeliveryStatus
1	7	24	Transaction	Cash Deposit Successful	Cash deposit of Rs. 5000.00 completed successfully...	SMS	2026-08-14 21:39:57.3323330	Sent
2	8	52	Transaction	ATM Deposit Successful	ATM deposit of Rs. 80000.00 completed successfully...	SMS	2026-08-14 21:39:57.3343240	Sent
3	9	19	Transaction	Bank Charge Applied	Bank charge of Rs. 80000.00 has been applied. Ref...	SMS	2026-08-14 21:55:13.6619957	Sent
4	10	39	Transaction	ATM Withdrawal Successful	ATM withdrawal of Rs. 37000.00 completed success...	SMS	2026-08-14 21:55:13.6643520	Sent
5	11	24	Transaction	Fixed Deposit Booked	Fixed Deposit of Rs. 20000.00 booked successfully...	SMS	2026-08-15 12:43:33.9150613	Sent
6	12	42	Transaction	Fixed Deposit Booked	Fixed Deposit of Rs. 39000.00 booked successfully...	SMS	2026-08-15 12:43:33.9488762	Sent
7	17	35	Transaction	Transaction Successful	Transaction of Rs. 5000.00 completed successfully. ...	SMS	2026-08-15 19:52:54.4472898	Sent
8	18	16	Transaction	Transaction Successful	Transaction of Rs. 10000.00 completed successfully...	SMS	2026-08-15 19:52:54.4492804	Sent
9	21	15	Transaction	UPI Transfer Successful	UPI transfer of Rs. 2000.00 completed successfully...	SMS	2026-08-16 12:33:03.8058365	Sent
10	22	32	Transaction	UPI Transfer Successful	Rs. 2000.00 received through UPI. Reference: TRF-2...	SMS	2026-08-16 12:33:03.8068341	Sent
11	23	39	Transaction	IMPS Transfer Successful	IMPS transfer of Rs. 10000.00 completed successful...	SMS	2026-08-16 12:33:03.8078310	Sent
12	24	30	Transaction	IMPS Transfer Successful	Rs. 10000.00 received through IMPS. Reference: TR...	SMS	2026-08-16 12:33:03.8088363	Sent
13	25	41	Transaction	NEFT Transfer Successful	NEFT transfer of Rs. 25000.00 completed successful...	SMS	2026-08-16 12:33:03.8098241	Sent
14	26	53	Transaction	NEFT Transfer Successful	Rs. 25000.00 received through NEFT. Reference: TR...	SMS	2026-08-16 12:33:03.8108230	Sent
15	27	31	Transaction	Transaction Successful	Transaction of Rs. 47851.28 completed successfully...	SMS	2026-08-16 18:54:48.4516698	Sent
16	28	31	Transaction	Transaction Successful	Transaction of Rs. 47851.28 completed successfully...	SMS	2026-08-16 18:54:48.4561864	Sent
17	29	31	Transaction	Transaction Successful	Transaction of Rs. 47851.28 completed successfully...	SMS	2026-08-16 18:54:48.4572174	Sent

Query executed successfully.

SATYAM\SQLEXPRESS (16.0 RTM) | SATYAM\ABHI (53) | BankingSystem | 00:00:00 | 17 rows